

Computational Discovery of New Fungicide Candidates to Protect Potato Crops

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Introduction

- Potato production is destroyed worldwide by fungal pathogens, leading to billions of dollars in losses annually
- Fungi are rapidly developing multi-drug resistance to existing fungicides
- New fungicides with novel modes of action are therefore critical
- YeF3 is critical to cell function, yet only exists in the fungi kingdom making it the perfect target
- Therefore, targeting YeF3 leads to highly effective, non-toxic eradication of fungi

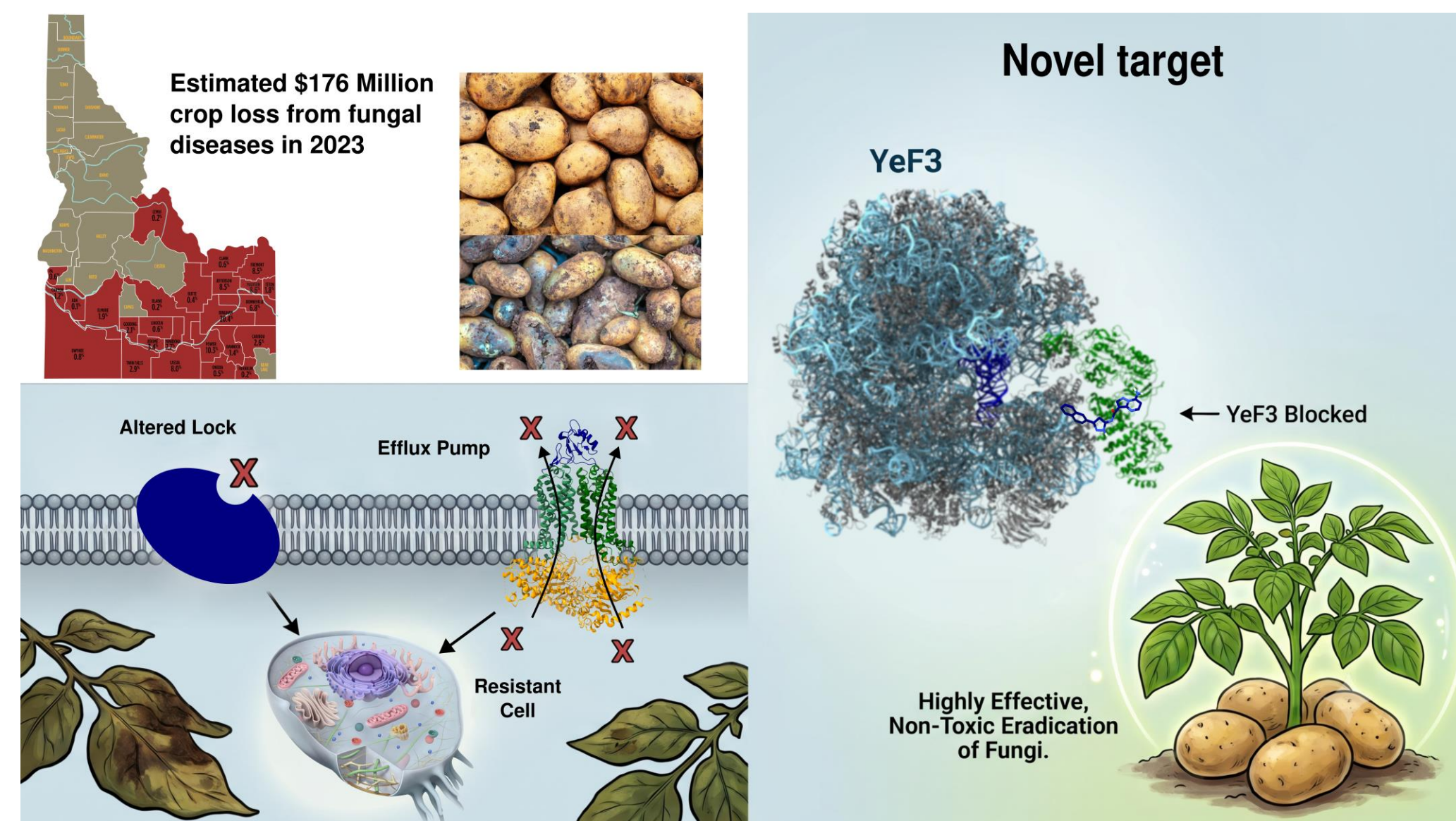


Figure 1: Infographic of Idaho economic crop loss, common methods of fungal resistance, and YeF3 as a novel target for antifungals

Workflow

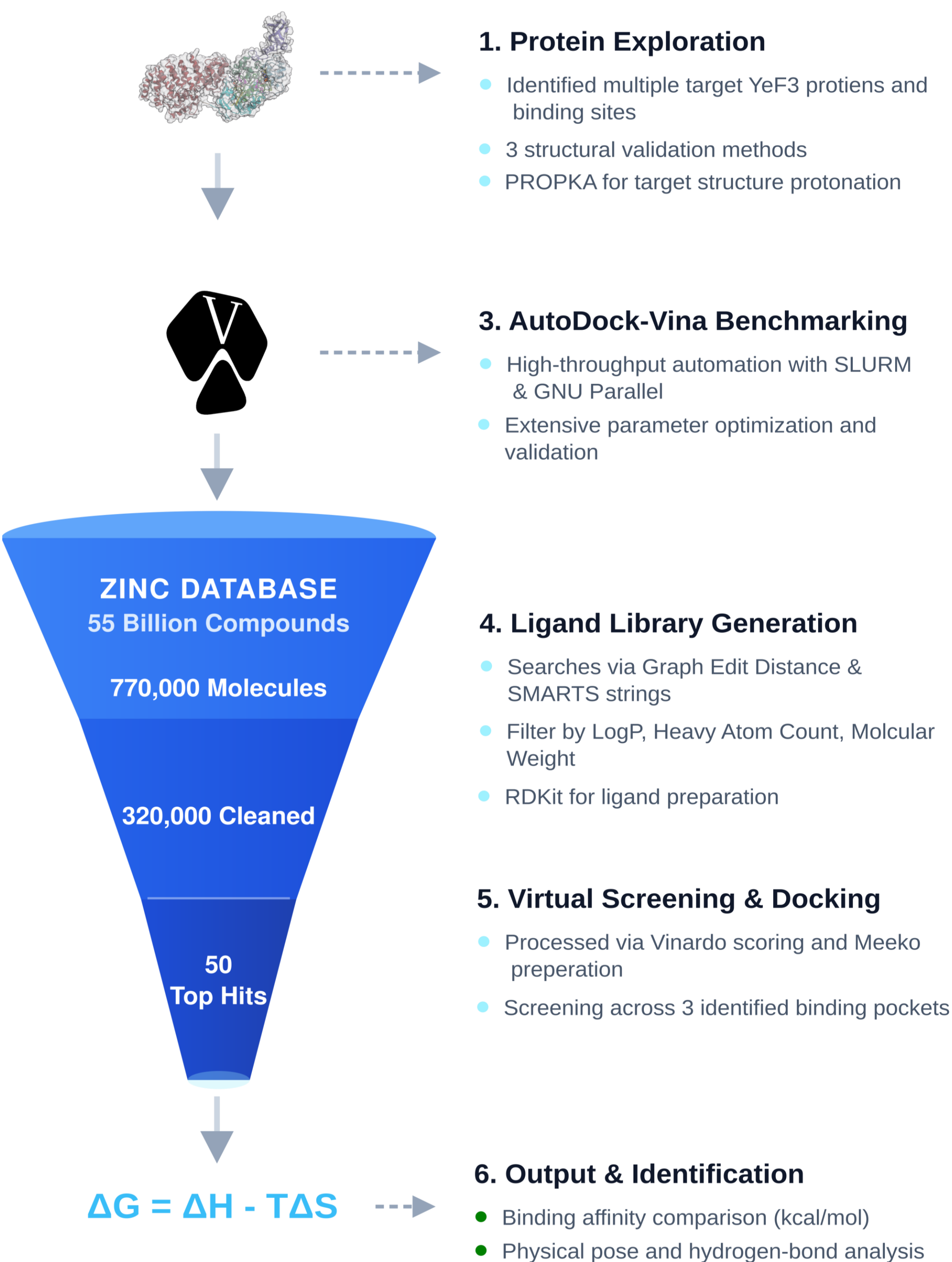


Figure 2: Methods image of pipeline for High Throughput Virtual Screening

Mechanism of Action

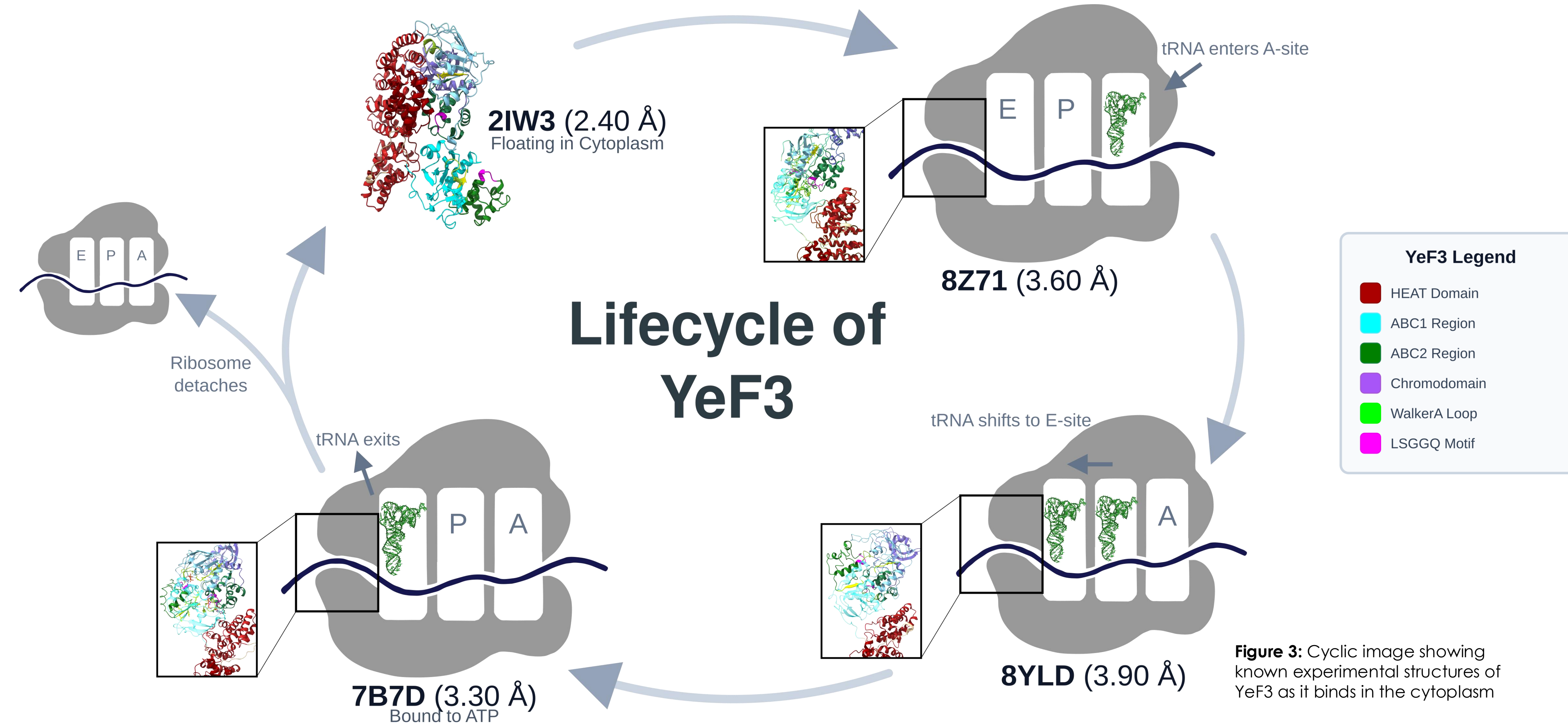


Figure 3: Cyclic image showing known experimental structures of YeF3 as it binds in the cytoplasm

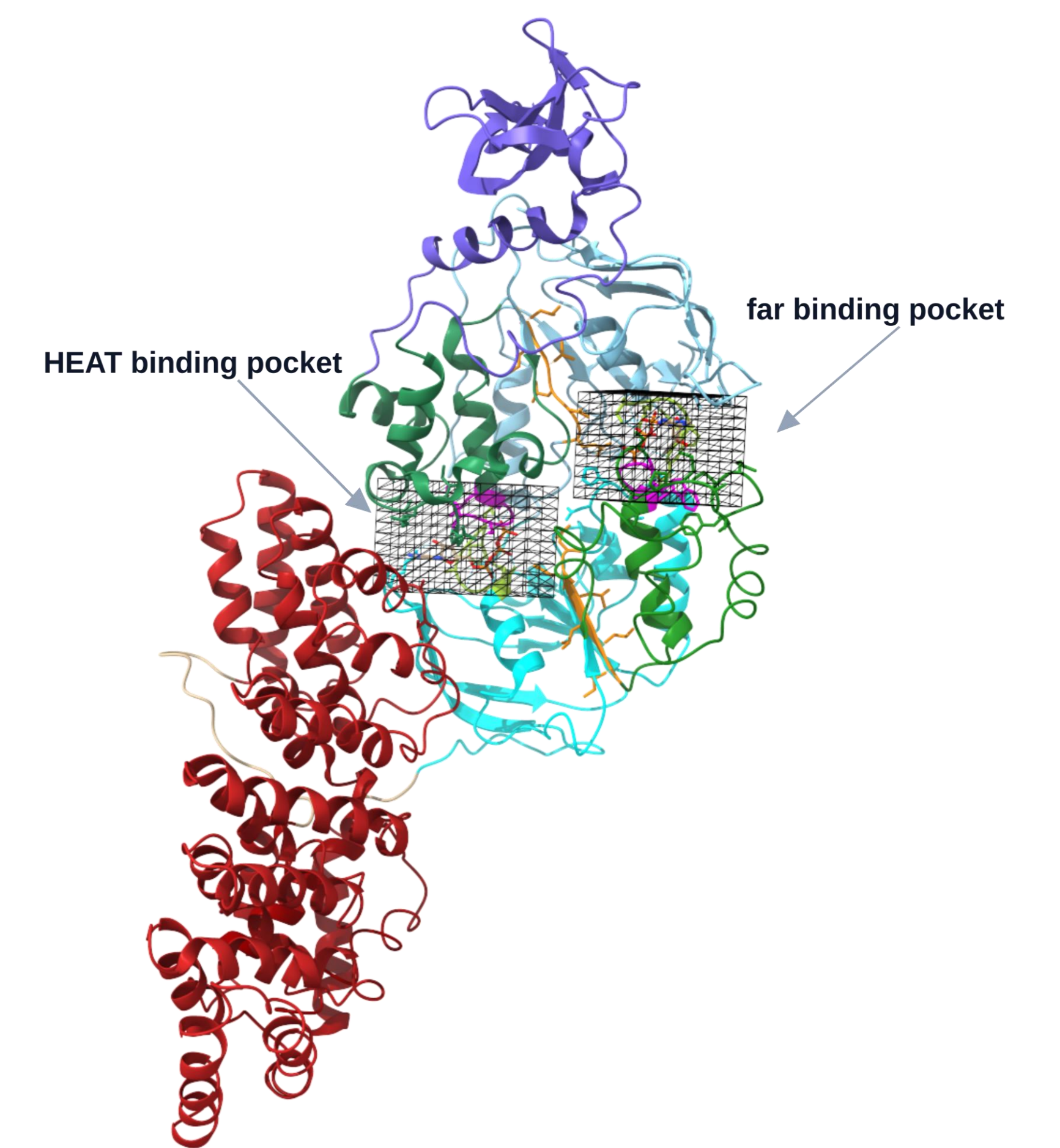


Figure 4: Diagram showing Autodock Vina grid boxes around two binding pockets in 7B7D experimental structure.

Results

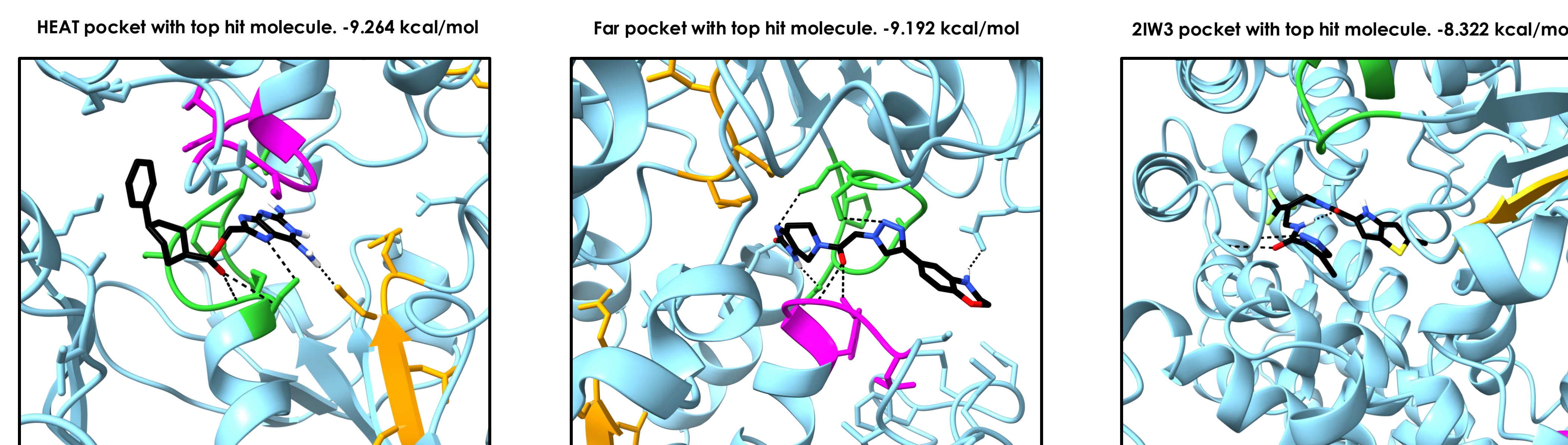


Figure 5: Poses of representative top hit molecule in each pocket with hydrogen bonds via ChimeraX

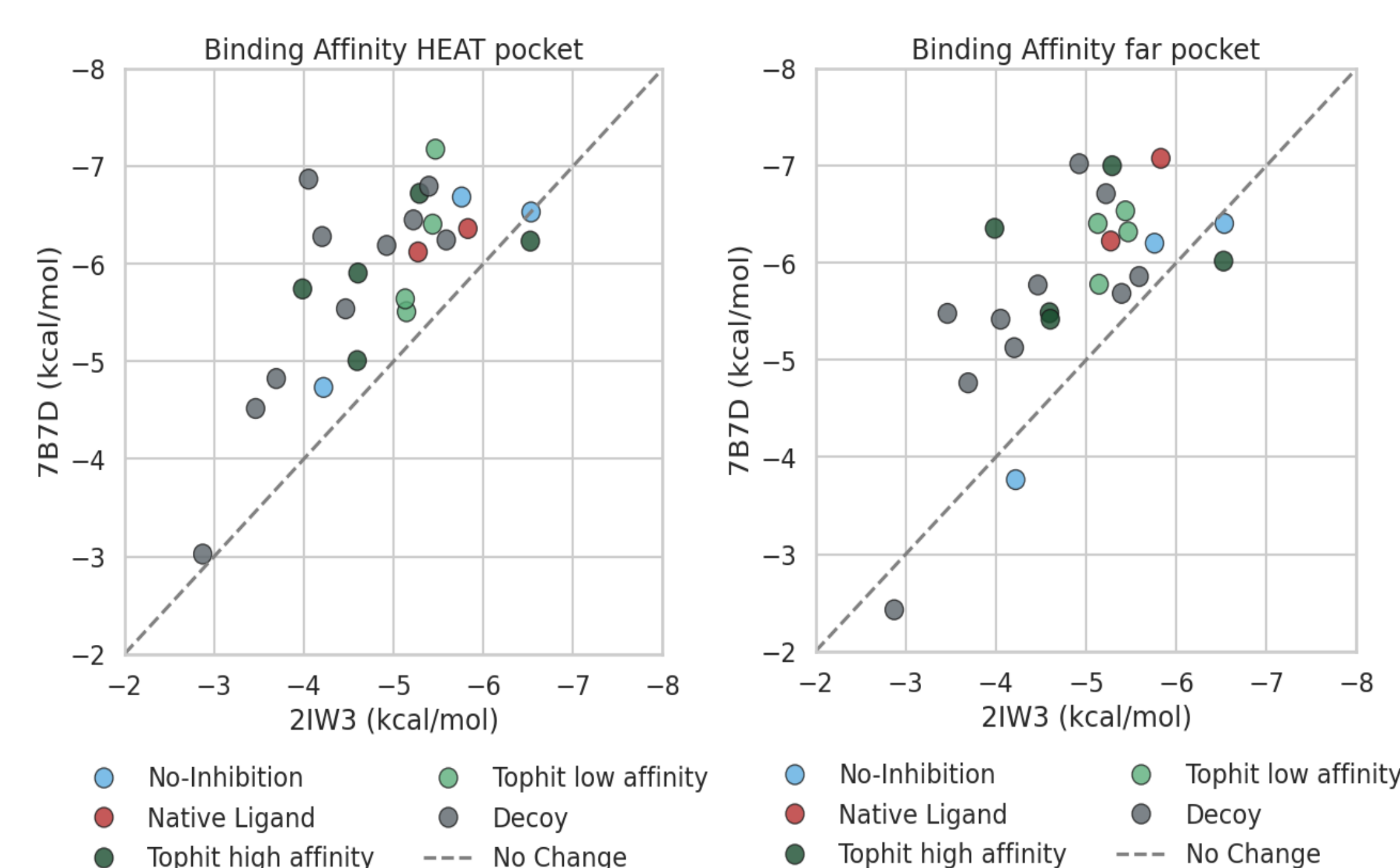


Figure 6: Binding affinity of 2IW3 binding site benchmarked against 7B7D pockets

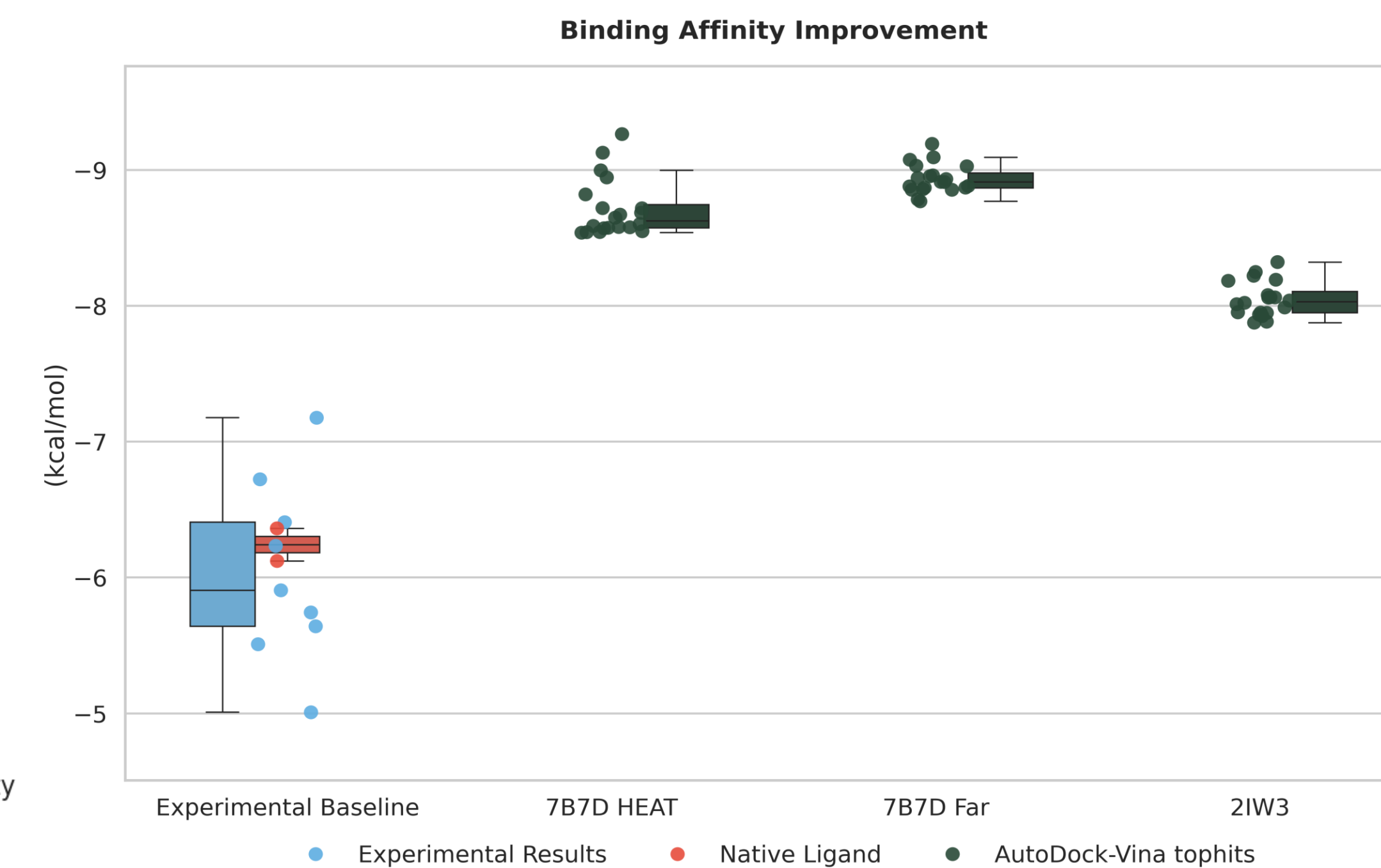


Figure 7: Comparison of top hit experimental binding affinities from high throughput virtual screening showing an improvement of 3 kcal/mol

Conclusion

Our virtual screening identified several promising compounds with strong predicted binding to the fungal protein YeF3, particularly in the 7B7D conformation. Many of the top candidates bind within the highly conserved ABC-binding cassette motifs, suggesting they may be less susceptible to resistance-causing mutations. The stronger predicted binding observed in the 7B7D pockets highlights these sites as promising targets for future optimization and experimental validation. Notably, several compounds improved predicted binding affinity by up to 3 kcal/mol compared with reference molecules, representing a substantial gain in potential binding strength.

Future Work

- **Benchmark and test the performance of DOCK6** that utilizes anchor-and-grow algorithm.
- **Prioritize and compare top hits** across binding pockets through consensus docking and structural similarity analysis.
- **Evaluate drug-like properties** of the highest-ranked compounds using SwissADME, including molecular weight, LogP, heavy atom count, and other ADME metrics.
- **Refine predicted binding poses** using AI-based structure prediction methods such as Boltz-2 and REINVENT4.
- **Experimentally validate** the most promising candidates to confirm their inhibitory activity against YeF3.
- **Optimize confirmed hits** through rational drug design and focused analog library generation to improve potency and drug-like properties.

References

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